

Research overview

Sean Muller · October 2026

Independent quantitative research in systematic equities: the question, how results are tested, what the evidence shows and where it stops.

EVERY PERFORMANCE FIGURE IN THIS BRIEF IS A HISTORICAL BACKTEST. NO LIVE OR PAPER-TRADING RESULTS ARE SHOWN.

The research

I study cross-sectional and relative-value behavior in U.S. equities as an empirical question. The main question: whether relative-value effects among U.S. single stocks are worth trading after realistic costs, and how much of an apparent result survives testing. A stock that moves away from comparable stocks may be under temporary price pressure or repriced for a reason that lasts; the two are hard to tell apart when the move happens, and costs decide whether the difference is worth trading.

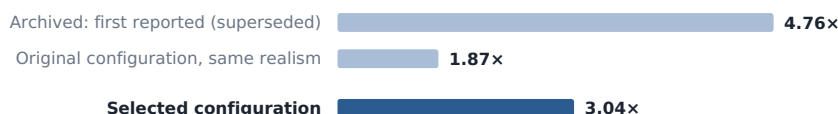
How a result is tested

A test is designed before its result is read: the baseline it has to beat, the placebo that has to fail, and the outcome that would count against the idea. Most of the work is then trying to break the result. Three examples:

- **Evaluation design.** A repeatability score read 0.48 on a random split and 0.02 when the same data were split by blocks of time.
- **Multiple testing and selection bias.** A Sharpe ratio had been deflated for an assumed 100 trials. A recount of the record found about 540 runs.
- **Placebo controls, power and feasibility.** Planted effects of two realistic sizes were found only about a fifth and about a third of the time.

What the evidence shows

In late September 2026 I reported a two-year walk-forward backtest that ended at 4.76×. An audit I directed in October found hindsight in its list of tradable stocks, and its costs were a flat charge with every order filled in full. Rebuilt on a point-in-time universe with modeled fills, the original configuration ends at 1.87×. A less impressive but correctly specified backtest is more valuable than an impressive contaminated one.



Ending value of \$1 after about two years in three historical backtests. Bars are drawn from 1×, the starting dollar. The last bar is not a further audit step: its liquidity requirement was chosen afterward, on the same window, so it is not independent out-of-sample evidence.

The selected configuration keeps both corrections and revises the liquidity screen and requirement. It ends at 3.04× (about 70% a year), with a Sharpe ratio of 1.84 on daily returns, no risk-free rate subtracted, a maximum drawdown of −19.1% and 780 simulated trades.

Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85. The book can use leverage and no financing cost is charged. Over the same window the S&P 500 rose about 42% on a price basis: context, not a like-for-like comparison. Factor and benchmark decomposition remain part of the ongoing evidence-extension program used to characterize where the historical performance originates.

Selected case studies

- Restating a backtest whose stock universe was built with hindsight.
- Testing my own hypothesis about crowded chart triggers until no trading claim was left.
- Selecting on a noisy proxy: a result that looked real and was measurement error.

Research infrastructure

Built because the research needed it: minute-resolution history for about 5,400 U.S. equities, a walk-forward research replay, a transaction-cost model, a ledger of 500+ documented experiments, and a trading engine. The platform has progressed from research and simulation into live trading with real capital through broker-integrated execution.

Limits

- The headline is one implementation of one two-year window, and its configuration was chosen on that window, so it is not independent out-of-sample evidence; intervals, start-date sensitivity and factor decomposition for it are being extended as part of the continuing robustness program.
- A small fraction of trades accounts for more than half of net profit, and a two-year compounding multiple is sensitive to small changes.
- Costs are modeled; execution data from live operation are being accumulated to calibrate them. The point-in-time reconstruction is still being completed, and the work has not yet had outside review.

About

I am an undergraduate at the University of Miami, graduating in December 2026 with a B.A. in Mathematics and a B.S. in Finance & Accounting. I am looking for full-time, new-graduate roles in quantitative research and quantitative trading from December 2026. The research questions, methodology, system design, and interpretation are independently directed; implementation uses extensive AI-assisted development.

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